SUMMARY · MK 2 FINANCIAL MANAGEMENT & STRATEGY · WEEK 2

Financial Statement & Ratio Analysis

Week 1 taught reading statements; Week 2 teaches analyzing them. Reporting supplies information; analysis turns it into judgment. You learn a 6-step framework, common-size analysis, five ratio families and DuPont.

Material

1 Reporting vs analysis

Reporting supplies information; **analysis** assesses past and likely future performance for capital-allocation decisions. Analysts ask whether results beat expectations, what drove them, whether margins last and what could reverse them.

Read the **notes** (where real analysis often begins), management commentary (critically) and the audit report (fair presentation, not a guarantee of success). Use top-down logic: macro → industry → company.

2 Earnings quality: profit isn't always performance

Reported profit can diverge from economics via omitted value, method choices, estimates and earnings management. Ask: sustainable, cash-backed, recurring, consistent with fundamentals?

3 6-step framework and tools

1. Purpose
2. Data
3. Process

4. Analyze
5. Conclude
6. Follow up

Computation isn't analysis — a ratio is an indicator. Tools: graphics, regression, common-size (vertical: item ÷ base; horizontal: current ÷ base – 1), ratios.

Red flag: revenue +19%, receivables +38%, inventory +58% → loose credit, slow collection, aggressive recognition, obsolete stock or channel stuffing. Communicate starting from the decision question.

4 Five ratio families

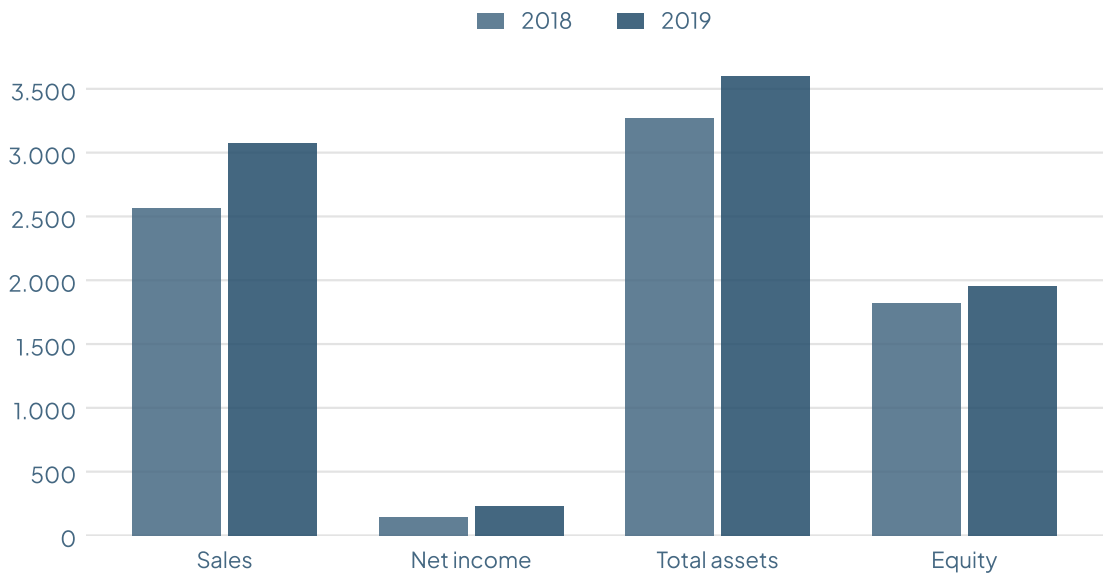
Users: shareholders, creditors, management. Compare via trend, cross-section, benchmarking; beware industry, size, seasonality, policies, one-offs.

Liquidity: current ratio, quick ratio — too high may mean idle capital. **Activity:** inventory turnover, collection period, asset turnover. **Solvency:** debt ratio, D/E, interest coverage — leverage lifts ROE and risk. **Profitability:** gross/operating/net margin, ROA, ROE. **Valuation:** P/E, P/CF, P/S, P/BV. The families are linked.

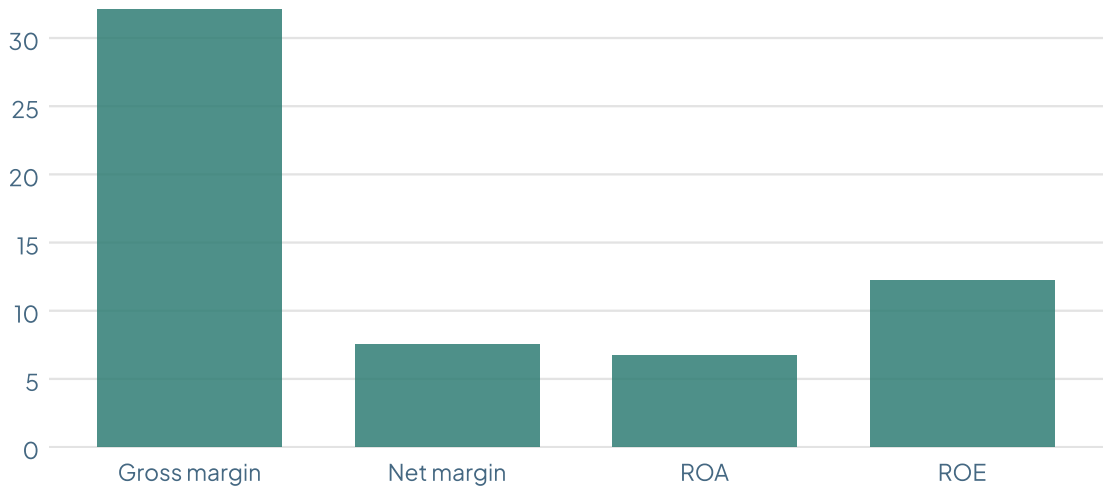
5 Bartlett's ratios charted

Deep dive

Bartlett: 2018 vs 2019 ($000)



Bartlett profitability 2019 (%)



1.97x
Current ratio

1.51x
Quick ratio

45.7%
Debt ratio

7.09x
Inventory turnover (~51 days)

Quick read

Sales +20%, net income +56% → margin up from 5.8% to 7.5%.

6 DuPont analysis: decomposing ROE

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ROE = Net margin × Asset turnover × Financial leverage
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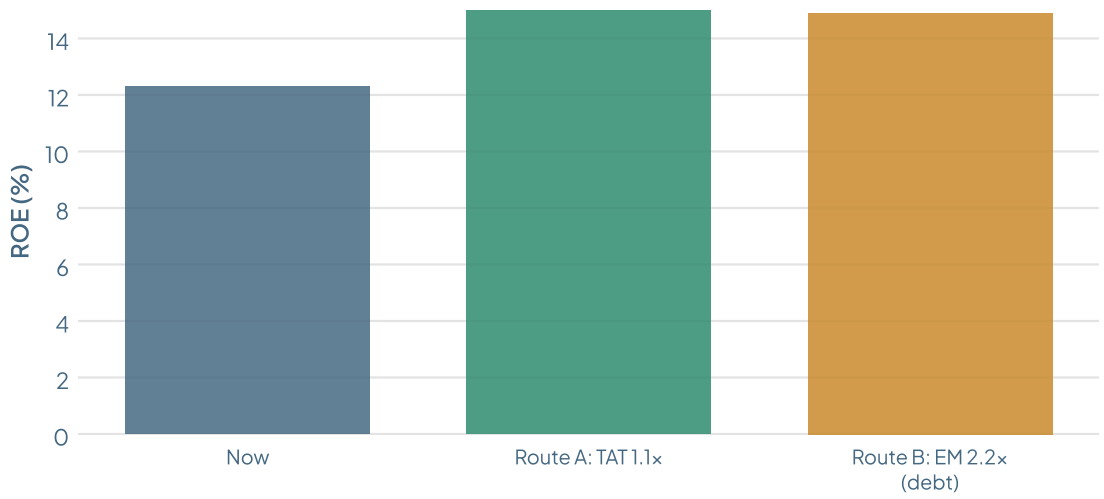
	A	B	C
ROE	20%	20%	20%
Margin	10%	5%	3%
Turnover	2.0×	2.0×	4.45×
Leverage	1.0×	2.0×	1.5×

A wins on margin, C on efficiency, B relies on debt — high ROE driven by leverage is fragile. Signals → actions: falling margin (pricing, COGS), falling turnover (capacity, inventory, collections), rising leverage (cash flow, covenants), rising ROE (decompose before celebrating). IFSA's 5-way DuPont splits margin into tax burden, interest burden and EBIT margin.

7 Try it: the DuPont calculator Deep dive

Defaults are Bartlett's (NPM 7.5%, TAT 0.90×, EM 1.82× → ROE ≈ 12.2%). Which route to 15% ROE is healthier?

Two routes to ~15% ROE



A: 7.5% × 1.1 × 1.82; B: 7.5% × 0.9 × 2.2. Same result, different risk.

! **High ROE isn't always good**
Debt-driven ROE is fragile.

8 Ratios in forecasting (bridge to Week 3)

Historical ratios drive forecast models (sales growth, margins, receivable/inventory/payable days, leverage). Don't extrapolate mechanically — combine with strategy, industry and macro assumptions.

9 The four financial statements as one system

Before computing ratios, the slides stress **statement literacy**: every ratio is only as good as the statements behind it. Five parts must be read together:

Statement	Main question	How to remember it
Statement of financial position (balance sheet)	What does the company own, what does it owe, and what is left for the owners?	A photo on one date. Assets = Liabilities + Equity

Statement of comprehensive income	Did the company generate profit (and other comprehensive income, OCI)?	A film over one period. Comprehensive income = Net income + OCI
Statement of changes in equity	How did owners' claims change?	The bridge: profit comes in, dividends go out, paid-in capital changes
Statement of cash flows	Where did cash come from and where did it go?	Operating, investing, financing. Ending cash = beginning cash + CFO + CFI + CFF
Notes to the statements	Which policies, estimates, risks and details explain the numbers?	This is often where the real analysis begins

Net income (income statement) Retained earnings & dividends (changes in equity)

Equity on the balance sheet Cash reconciliation (cash flow statement)

≡ **Example of the links**

A credit sale of IDR 100 million today immediately raises *revenue* and *receivables*, but cash has not increased yet. Next month, when the customer pays, cash rises and receivables fall — revenue does not change again. Understanding this **timing** stops us from treating profit as if it were cash.

What to scan first on the balance sheet: current assets (can they cover near-term needs?), non-current assets (what asset base is needed to generate revenue?), current liabilities (what must be paid soon?), non-current liabilities (how much long-term obligation is embedded?), and equity (how much of the assets is financed by owners?).

Three supporting documents: (1) the *notes* — accounting policies, estimates, contingencies, segments, debt maturities; (2) *management commentary* — the narrative about strategy and outlook, useful but to be read critically; (3) the *audit report* — an independent opinion that the statements are fairly presented under the standards, **not** a guarantee that the business will succeed.

Sources beyond the financial statements: annual and interim reports, investor presentations, stock exchange and regulator filings, prospectuses, industry and macro data, credit ratings, news, and direct observation (site visits, talking to customers and suppliers). The logic is *top-down*: understand the macro economy → industry prospects → then judge the company's strategy and results in that context. Financial statements tell you *what* changed; wider evidence explains *why* and whether the change can last.

10 From ratio signal to action, and how to compare

The slides close the ratio chapter with one message: **"computation is not analysis"**. A ratio is only an indicator; analysis means explaining why the number looks that way and what action follows.

Three ways to compare:

Trend (time-series) analysis The same company across years — to see improvement, deterioration or volatility.

Cross-sectional Several firms at the same point in time — comparing performance with competitors.

BenchmarkingAgainst the best competitor or the industry average — to see the gap you want to close.

Be careful
Ratios are shaped by the industry's business model, firm size, seasonality, accounting policies, life-cycle stage and one-off events. A current ratio of 1.2 can be healthy in retail (cash turns over fast) but thin for a construction company.

Signal	Possible diagnosis	Action / next evidence to look for
Margin falling	Price pressure, cost inflation, product mix, weak expense control	Review pricing, COGS detail, segment margins, supplier contracts

Asset turnover falling	Assets not generating sales, inventory build-up, slower collections	Review capacity, inventory ageing, credit policy, capex plan
Leverage rising	Debt-financed growth, working-capital strain, buybacks/dividends	Review cash flow, covenants, maturity schedule, interest coverage
ROE rising	Could be real improvement, could be higher risk	Decompose with DuPont before celebrating

Short answers to the discussion questions on the slides:

- *Which statement would a lender read first?* Cash flows and the balance sheet — creditors care about the ability to pay interest and principal. An equity investor usually starts with the income statement (growth and margins), then the cash flow statement to test earnings quality.
- *Why can a profitable company face liquidity problems?* Because profit is recorded when the sale happens, while cash only arrives when it is collected. If receivables and inventory grow fast, cash is absorbed into working capital.
- *Current ratio of 4.0 or 1.5?* It depends. 4.0 is safe but may mean idle cash or obsolete inventory. Ask: what makes up current assets, how old are inventory and receivables, and what is the industry average?
- *Can ROE rise while the firm becomes riskier?* Yes — if the rise comes from leverage (the equity multiplier) rather than margin or turnover. DuPont shows this.
- *Which ratio would you trust least without reading the notes?* Earnings-based ratios (margins, P/E) — because they depend on depreciation method, revenue recognition and reserve estimates.

Key takeaways

1. Reporting = information; analysis = judgment for capital allocation.
2. The notes are often where analysis really begins.
3. Computing isn't analyzing — ratios need context.
4. Vertical/horizontal common-size; watch receivables/inventory outgrowing sales.
5. Five ratio families: liquidity, activity, solvency, profitability, valuation.
6. DuPont: ROE = margin × turnover × leverage; leverage-driven ROE is fragile.
7. Forecasting uses historical ratios plus strategy, industry and macro judgment.

Case studies

Case & worked answers

Bartlett Company: one-page diagnosis

Data ($000): sales 3,074 (2,567); GP 986; NI 231 (148); CA 1,223; CL 620; TA 3,597 (3,270); TL 1,643; equity 1,954 (1,820).

CR 1.97×, QR 1.51×, inventory turnover 7.09×, debt ratio 45.7%, gross margin 32.1%, net margin 7.5%, ROA 6.7%, ROE 12.2%. DuPont: 7.5% × 0.90 × 1.82 ≈ 12.2%.

Diagnosis: sales +~20%, NI +~56% → better margins; healthy liquidity; moderate leverage. ROE is margin-driven, not debt-driven — good quality; asset turnover (0.9×) is the improvement area. Ask management what drove margins and whether new assets are fully productive; benchmark against industry and check operating cash flow.

Discussion: lenders read cash flow and balance sheet first; equity investors income and cash flow; profitable firms can be illiquid when profit is tied up in

receivables/inventory; ROE can rise with risk via leverage; EPS/P-E and inventory turnover need the notes.

Worked examples

Problem 1 Easy **Likuiditas**

Aset lancar Rp600 juta terdiri dari kas 80, surat berharga 40, piutang 180, dan persediaan 300. Liabilitas lancar Rp400 juta. Hitung current ratio dan quick ratio, lalu jelaskan artinya.

1. `Current ratio = Aset lancar ÷ Liabilitas lancar = 600 ÷ 400 = 1,5`
2. `Quick ratio = (Kas + Surat berharga + Piutang) ÷ Liabilitas lancar = (80 + 40 + 180) ÷ 400 = 0,75`
3. Bandingkan: current ratio tampak sehat (1,5×), tetapi separuh aset lancar adalah persediaan. Tanpa persediaan, perusahaan hanya punya Rp0,75 untuk setiap Rp1 utang jangka pendek.

Answer: Current ratio 1,5×; quick ratio 0,75×. Likuiditas bergantung pada kemampuan menjual persediaan.

Problem 2 Medium **DuPont**

Penjualan Rp2.000 juta, laba bersih Rp150 juta, total aset Rp1.600 juta, ekuitas Rp800 juta. Pecah ROE dengan DuPont.

1. Net profit margin = 150 ÷ 2.000 = **7,5%**.
2. Perputaran aset (TAT) = 2.000 ÷ 1.600 = **1,25×**.
3. Equity multiplier (leverage) = 1.600 ÷ 800 = **2,0×**.
4.
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ROE = 7,5% × 1,25 × 2,0 = 18,75%
```
Cek langsung: 150 ÷ 800 = 18,75% ✓
5. ROA = 7,5% × 1,25 = 9,375%. Leverage menggandakan ROA menjadi ROE.

Answer: ROE 18,75% = margin 7,5% × perputaran 1,25 × leverage 2,0.

Problem 3 Easy **Solvabilitas**

EBIT Rp300 juta, beban bunga Rp60 juta, total liabilitas Rp800 juta, total aset Rp1.600 juta, ekuitas Rp800 juta. Hitung interest coverage, debt ratio, dan debt-to-equity.

1. Interest coverage = EBIT ÷ bunga = 300 ÷ 60 = **5×** (laba operasi 5 kali lipat bunga).
2. Debt ratio = 800 ÷ 1.600 = **50%**.
3. Debt-to-equity = 800 ÷ 800 = **1,0×**.

Answer: Coverage 5×, debt ratio 50%, D/E 1,0×.

Problem 4 Easy **Activity ratios**

One year's data (IDR million): sales 3,000; COGS 1,800; inventory 280 at the start and 320 at the end; year-end receivables 250; average total assets 2,400.

Compute (a) inventory turnover and days of inventory, (b) the average collection period, (c) total asset turnover.

1. Average inventory = (280 + 320) ÷ 2 = 300.

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Inventory turnover = COGS ÷ Average inventory = 1,800 ÷ 300 = 6×
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2. Days of inventory = 365 ÷ 6 ≈ **60.8 days** — goods sit in the warehouse about two months on average.

3. Daily sales = 3,000 ÷ 365 = 8.22.

```
Average collection period = Receivables ÷ Daily sales = 250 ÷ 8.22 ≈ 30.4 days
```

4.

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Total asset turnover = Sales ÷ Average total assets = 3,000 ÷ 2,400 = 1.25×
```

Answer: Inventory 6× (about 61 days), collection about 30 days, asset turnover 1.25×. Compare with last year and the industry before drawing conclusions.

Problem 5 Easy **Profitability ratios**

Sales IDR 3,000 million, gross profit 900, operating income 360, net income 210. Average total assets 2,400, average equity 1,200. Compute GPM, OPM, NPM, ROA and ROE.

1. Gross profit margin = 900 ÷ 3,000 = **30%**.
2. Operating profit margin = 360 ÷ 3,000 = **12%**.
3. Net profit margin = 210 ÷ 3,000 = **7%**.
4. ROA = Net income ÷ Average assets = 210 ÷ 2,400 = **8.75%**.
5. ROE = Net income ÷ Average equity = 210 ÷ 1,200 = **17.5%**. DuPont check: 7% × 1.25 × 2.0 = 17.5% ✓

Answer: GPM 30%, OPM 12%, NPM 7%, ROA 8.75%, ROE 17.5%.

Problem 6 Medium **Valuation ratios: EPS, P/E, P/BV**

Net income IDR 210 billion, preferred dividends IDR 10 billion, 400 million common shares outstanding. Share price IDR 6,000. Common equity IDR 1,200 billion. Compute EPS, P/E, book value per share and P/BV.

1. Earnings available to common = 210 − 10 = IDR 200 billion.

2. ```
   EPS = 200 billion ÷ 400 million shares = IDR 500
   ```

3. P/E = Price ÷ EPS = 6,000 ÷ 500 = **12×** — investors pay IDR 12 for every IDR 1 of earnings.

4. Book value per share = 1,200 billion ÷ 400 million = IDR 3,000. P/BV = 6,000 ÷ 3,000 = **2.0×**.

Answer: EPS IDR 500, P/E 12×, BV/share IDR 3,000, P/BV 2.0×.
   ```

Problem 7 Medium **Common-size and horizontal analysis**

Last year → this year (IDR billion): revenue 1,000 → 1,190; receivables 200 → 276; inventory 150 → 237. Compute (a) the growth of each item (horizontal), (b) receivables as % of revenue (vertical) in both years. What should raise suspicion?

1

```
Horizontal: Current period ÷ Base period - 1
```

Revenue 1,190 ÷ 1,000 – 1 = **19%**; receivables 276 ÷ 200 – 1 = **38%**; inventory 237 ÷ 150 – 1 = **58%**.

2

```
Vertical: Line item ÷ Base item (revenue)
```

Receivables/revenue: 200 ÷ 1,000 = 20% → 276 ÷ 1,190 = **23.2%**.

3 Receivables and inventory grow far faster than revenue — a *red flag* like the slide example. Possible causes: looser credit, slower collections, aggressive revenue recognition, or obsolete/piling inventory.

Answer: Revenue +19%, receivables +38%, inventory +58%; receivables rise from 20% to 23.2% of revenue → investigate.

Practice questions

Try answering first. The answer key is below each question.

1. NPM 7.5%, TAT 0.9×, EM 1.82×. ROE ≈ …

A. 6.8%
B. 12.3%
C. 15.0%
D. 8.2%

Answer: B. 6.75% × 1.82 ≈ 12.3%.

2. The quick ratio differs from the current ratio because it…

A. Adds fixed assets
B. Excludes inventory
C. Adds long-term debt
D. Uses net income

Answer: B. Inventory is least liquid.

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3. A profitable firm is short of cash. Most likely cause…

A. Margins too high
B. Accrual profit tied up in receivables and inventory
C. Too little debt
D. Zero tax

Answer: B. Profit ≠ cash.

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4. Creditors look first at…

A. EPS growth
B. Liquidity, leverage, cash flow
C. Share price
D. Headcount

Answer: B. They want to be repaid.

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5. ROE rose because the equity multiplier jumped. This signals…

A. Better operations
B. Higher financial risk from debt
C. Better margins
D. Faster inventory

Answer: B. Higher EM = more debt.

Sources

- -P3MD- MK 02 Week 2.pdf
- Bab yang dirangkum di deck: IFSA ch.1 & ch.7 (Robinson dkk., Financial Analysis 3rd ed.), Zutter & Smart ch.3